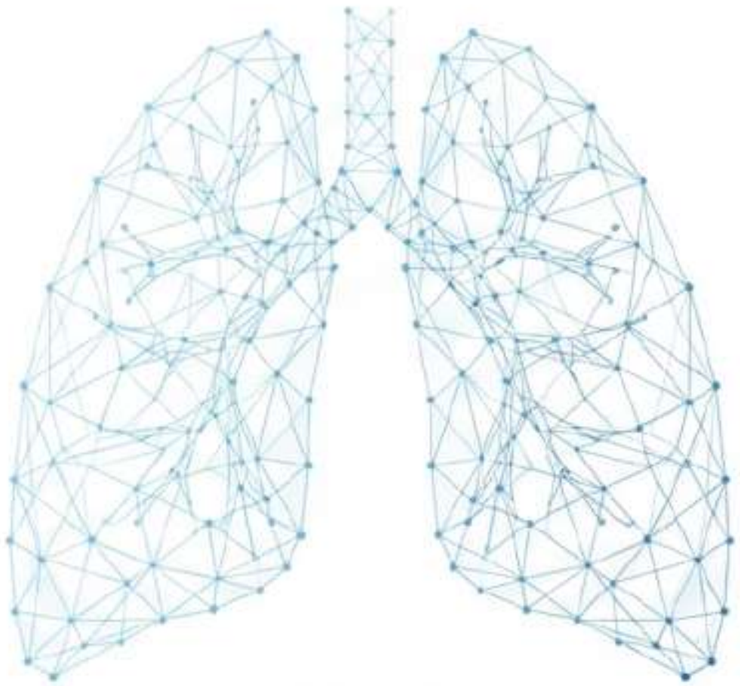


MedVision-Gemma

AI-Powered Clinical Decision-Making and Patient Guidance System

Prepared by: Esila Nur Demirci | Senior Engineer & Business Analyst

A Fusion of Engineering & Strategy

Esila Nur Demirci

Banking Security

Medical Empathy

12+ years of professional experience bridging complex' technical architectures with high-level business strategy.

Education

- Double Major in Mathematics & Computer Science (IKU),
- Master's in Information Technologies, ITU.

Industry Experience

- Track record of building robust, scalable systems for leading financial institutions (iş Bank, Akbank, TEB) and technology firms (Etiya),

"Building a robust, scalable, and secure system that doesn't just diagnose, but cares."

MedVision-Gemma

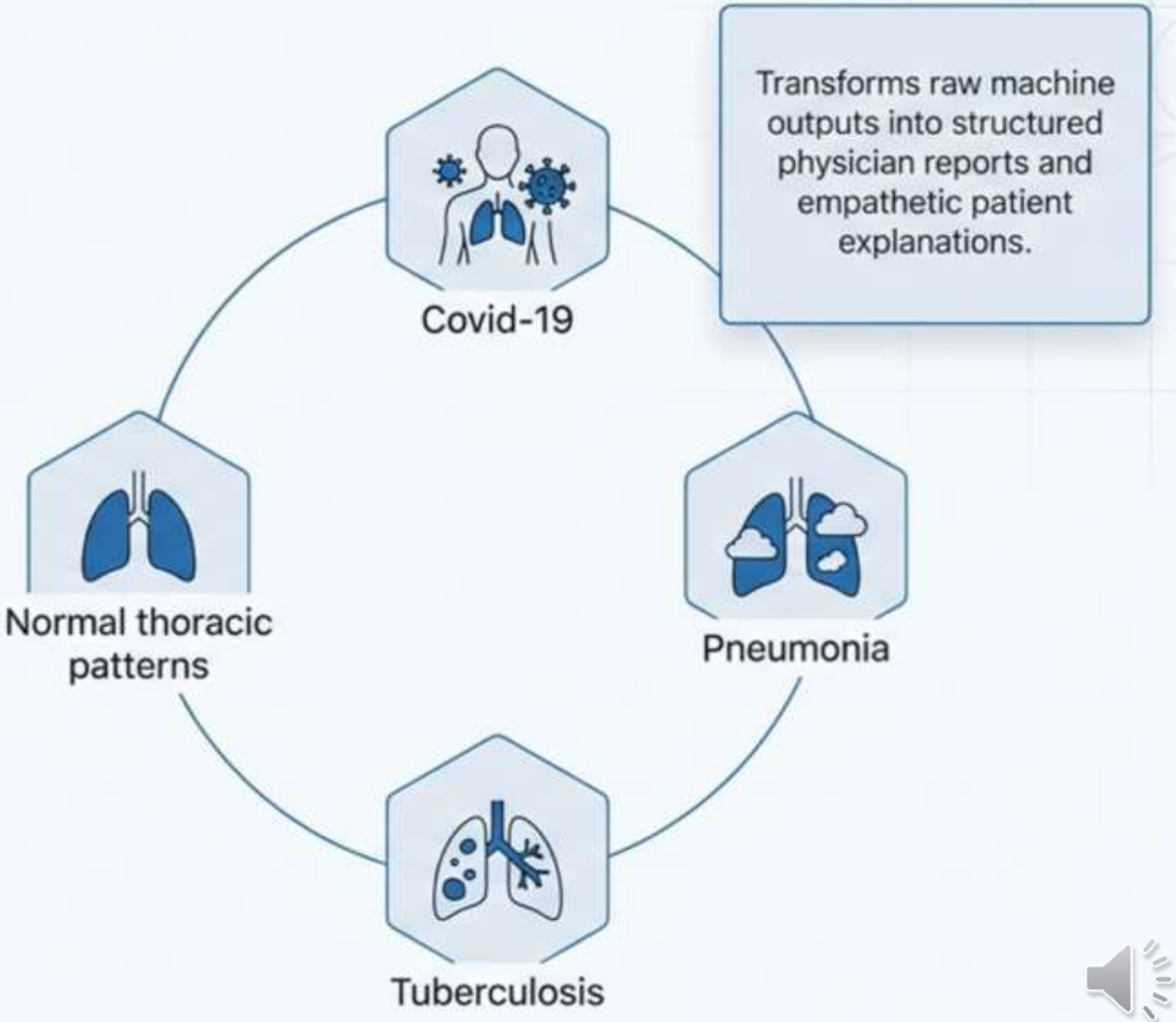
The Future of HAI-DEF: Where Clinical Precision Meets Human Empathy

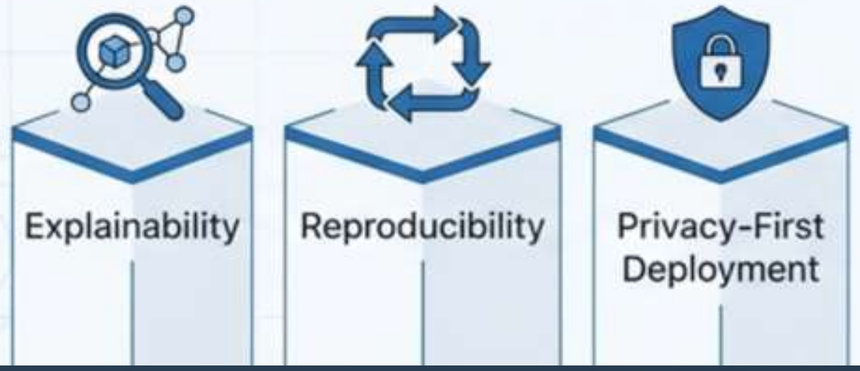
BEYOND PROBABILITIES. A UNIFIED CLINICAL DECISION SUPPORT SYSTEM.

Esila Nur Demirci | AI Software Engineer

High-Definition AI for Human-Centric Healthcare

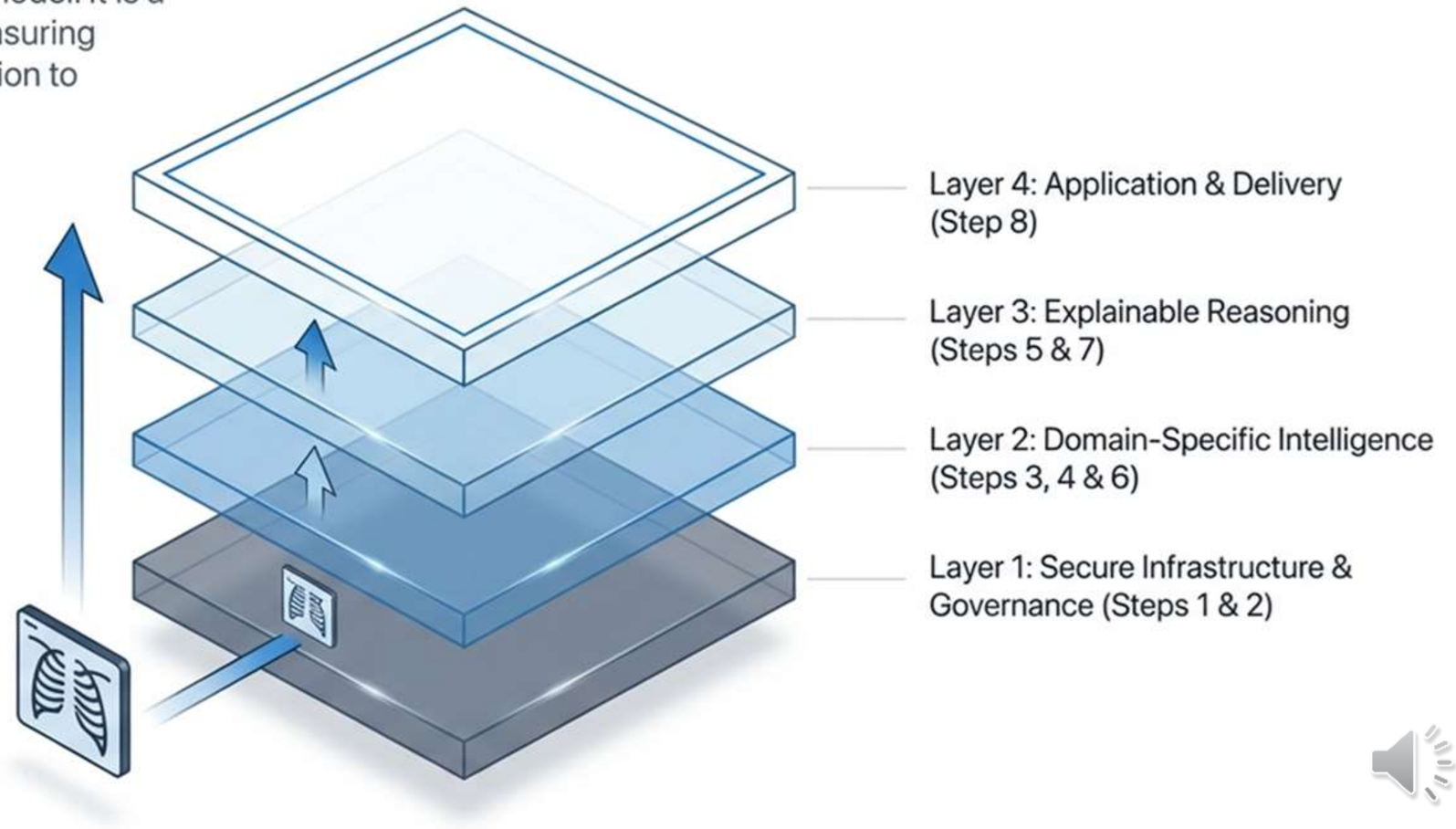
MedVision-Gemma is a diagnostic ecosystem designed to assist clinicians in detecting pulmonary conditions from chest X-ray images. Rather than delivering raw probabilities, the system leverages the High-Definition AI (HAI-DEF) framework to transform medical embeddings into tailored, actionable insights.





An 8-Step Ecosystem Built for Clinical Scalability

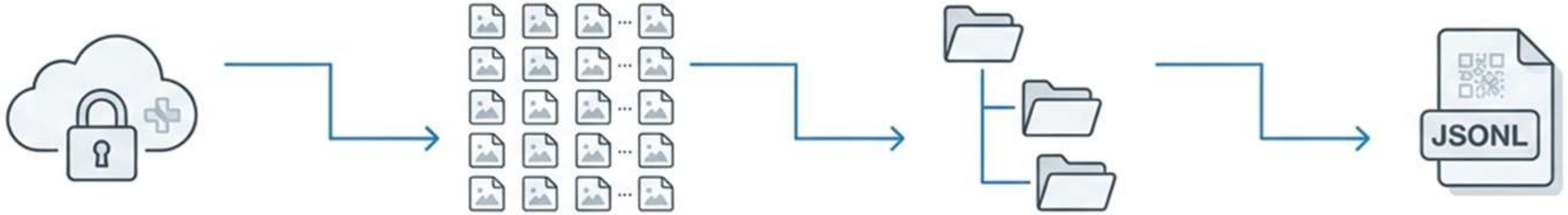
MedVision-Gemma is not a single model. It is a structured, deterministic pipeline ensuring reproducibility from raw data ingestion to patient delivery.



Establishing a Deterministic Data Foundation

Layer 1: Infrastructure

Before feature extraction, MedVision-Gemma establishes a fully governed data layer aligned with enterprise security standards. Manual labeling inconsistencies are eliminated through strict data lineage traceability.



1. Vertex AI (us-central1) with IAM-bound authentication

2. 72,297+ CXR images standardized into lossless PNG format

3. Deterministic folder hierarchy applied

4. Sealed JSONL manifest (cxr_batch.jsonl) generated

cxr_metadata.csv archived in Google Cloud Storage to preserve unbroken data lineage.

Extracting Subtle Pulmonary Manifestations

Layer 2: Intelligence

The pipeline transitions from raw pixels to medical-domain feature representations. Instead of generic vision encoders or the Google CXR-Foundation, the architecture leverages google/medsiglip-448 for specialized radiological learning.

Raw Chest X-Ray

MedSigLIP Encoder

1152-Dimensional Normalized Clinical Embedding

GPU-backed GCP VM (Tesla T4)

Atomic Parquet checkpointing (cxr_medsiglip_embeddings.parquet)

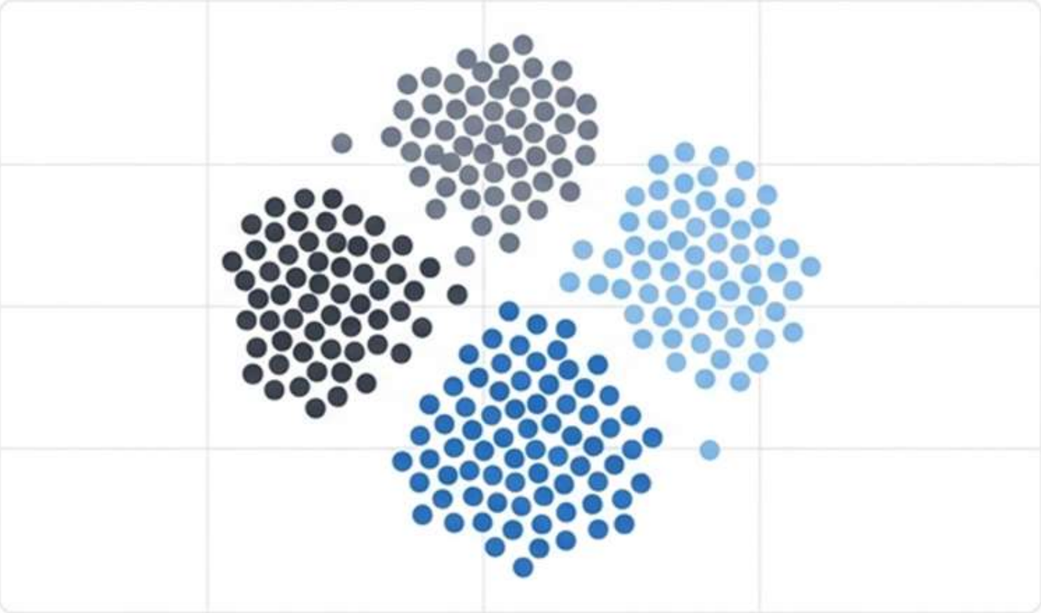
Resumable extraction logic avoids Colab instability, resulting in a strictly validated 1152-D feature space.

Sealing the Data Contract

Layer 2: Intelligence

Raw embeddings undergo strict validation to prevent leakage or schema drift, confirming that the embedding space reflects meaningful pathological separation rather than random projection noise.

- Strict 1152-D validation & Gatekeeper checks (no Unknown labels)
- Regex-based deterministic label extraction
- Stratified partitioning: 70% Train / 15% Validation / 15% Test



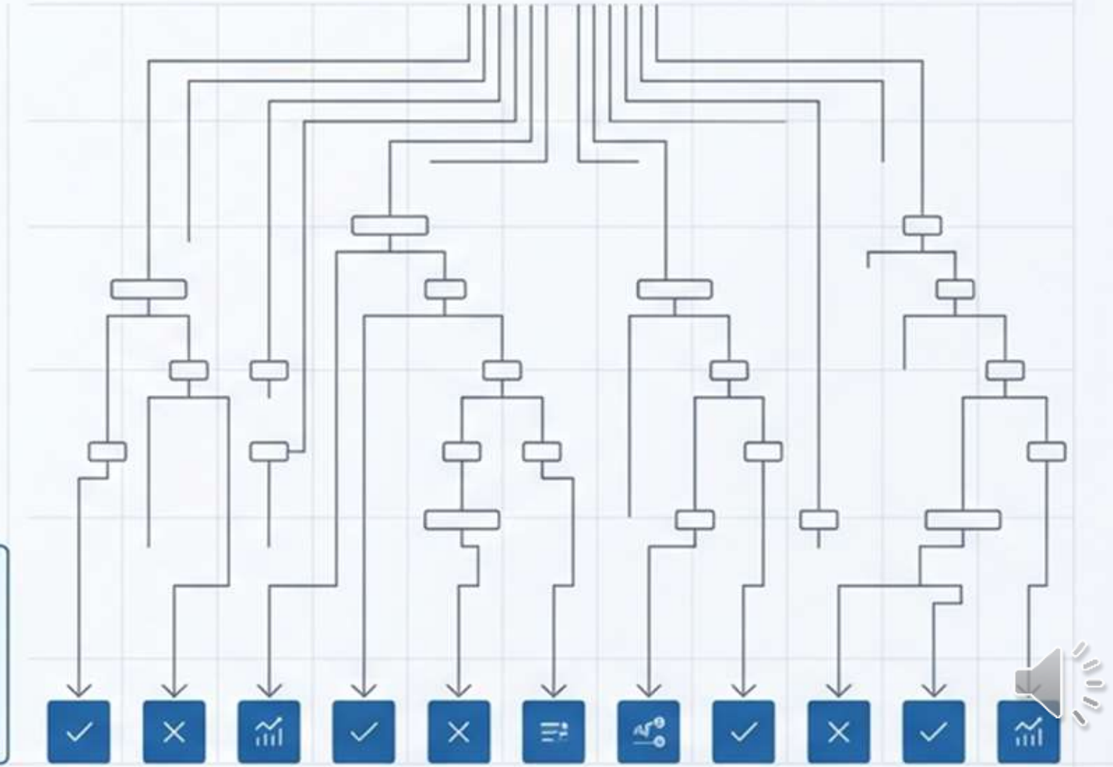
uri	label	embedding (1152-D)	split

Layer 3: Reasoning

Explainable Baseline Clinical Classification

The sealed dataset is converted into an interpretable intelligence system. Foundation embeddings are processed through a Random Forest classifier to ensure transparent decision signals.

- Transparent decision signals (Feature importance extraction) → Concerkince ability
- Robust under class imbalance (Balanced class weighting) → Depenseability
- Stable in high-dimensional spaces (max_features='sqrt') → Reproducible (Fixed random seed)



Clinicians require 'Why'. Random Forest preserves interpretability, transforming complex mathematical spaces into measurable feature importance.

Translating Machine Output into Clinical Intelligence

To fulfill the human-centric AI objective, the system evolves from a purely mathematical model into a structured intelligence interface using MedGemma.

Engineering highlight: RF provides the measurable feature importance; MedGemma translates it into non-prescriptive, governance-aligned structured reasoning.

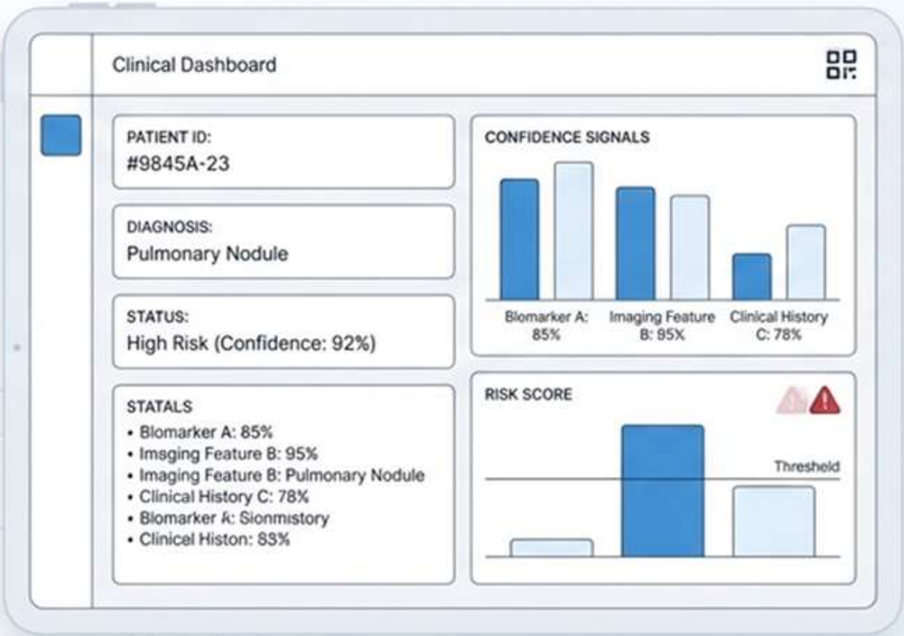
Bridging Machine Outputs with Human Care

Layer 4: Application

MedVision-Gemma translates raw data into two distinct, optimized experiences, ensuring precise communication for providers and empathetic guidance for patients.

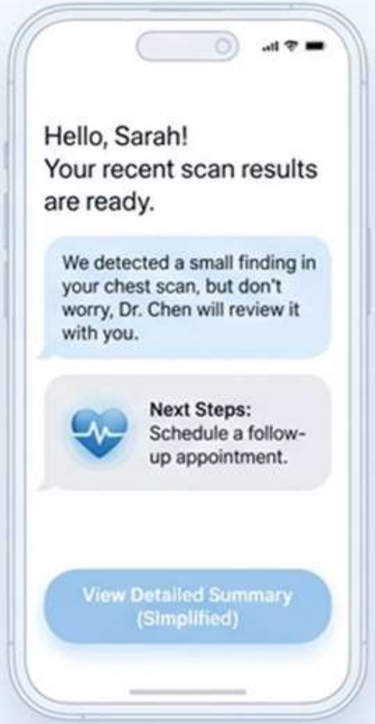
Physician-Facing Dashboard

Clinical, Precise



Patient-Facing Interface

Reassuring, Accessible



Extending the Ecosystem Beyond the Hospital

Layer 4: Application

To support continuous patient care, the project architecture includes a specialized Swift-based Health Companion application.

- Secure API integration
- HealthKit synchronization (Consent-based)
- Empathetic AI guidance generation
- Risk-awareness notifications
- Wellness reinforcement messaging

Transforms the project into a full-stack AI healthcare platform while strictly preserving data governance boundaries.

Privacy-First Deployment Strategy

Layer 4: Application

No raw patient data is ever exposed to external inference endpoints. The inference layer is containerized and modular, built to run securely across varying environments.

Tier 1: Research & Demo

Google Colab environment

Tier 2: Enterprise Cloud

GPU-backed Virtual Machine for stable, scalable processing

Tier 3: Hospital Secure

On-premise environment, fully contained behind hospital firewalls

Containerized microservices allow optional metadata fusion (age, symptoms, vitals) and future multimodal scaling without compromising privacy.

A Scalable Blueprint for Responsible Healthcare AI

MedVision-Gemma is not just a classifier. It is a structured, multi-layer AI ecosystem designed for real-world clinical scalability.

Deterministic Contracts
1152-D MedSigLIP embeddings & regex mapping

Explainable Decisions
Leakage-safe, Random Forest baseline

Modular Deployment
Container-ready, privacy-first infrastructure

Human-Centric Delivery
LLM-based structured report orchestration

Combining medical foundation intelligence with empathetic communication.

Conclusion: Excellence in Healthcare

MedVision-Gemma is not just an algorithm, it is a Healthcare Excellenc Framework.

- It combines Google's world-class medical infrastructure with local, secure, and explainable artificial intelligence.
- It offers a future where technology empowers doctors and patients are not left to face only a 'percentage'.